Increasing Value Capture by Enhancing Manufacturer Commitment—Designing a Value Cocreation System

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***Abstract*—*Designing a value cocreation system in the manufacturing industry has seen significant interest in the past two decades. The importance of a value cocreation system has increased along with the social expectations related to increasing resource efficiency and the emergence of enablers such as digitalization. In this article, I take this opportunity to disseminate some findings to industry and managers. Specifically, a review of product/service systems (PSSs) where a manufacturer—taking responsibility for and control of larger ranges in a value network—integrates their products and services to cocreate and capture higher value. PSS design implications and how to implement PSS design are discussed. Insights into designing a PSS in manufacturing practice are highlighted largely based on collaborative experience with large global manufacturers. Strategic management of design activities to avoid design fixation and service design trap is key to the successful design of PSSs.*

Key words: Circular economy (CE), manufacturing, product/service system (PSS), product-as-a-service (PaaS), systemic design

Why a Value Cocreation System in the Manufacturing Industry?

Manufacturers are facing increasing pressures concerning climate neutrality, resource efficiency, and other environmental impacts from their product lifecycles. The degree of pressure has never been so high in Europe and been increasing globally.

Companies are challenged to substantially change how they design, produce, and provide products. In tackling this challenge, services such as maintenance, repair, and remanufacturing provide a manufacturer with an excellent opportunity to decrease the environmental impacts while keeping or increasing competitiveness. Indeed, many manufacturers of complex products often provide related services to attract customers.

Providing services is inherently a cocreation activity with other actors, including product users. Providing both products and related services adds value to both the recipients and providers. Incorporating the value, more rational decision-making is supported in design. A system including both products and services can be identified as a design object. It is an enlarged system—a system of systems—for value cocreation that improves environmental performance and competitiveness.

These systems have become more prevalent; a system for product-as-a-service (PaaS) with washing machines for individual use in the circular economy (CE) context is one example (Sakao and Webster 2020).

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In improving outcomes, these systems require some rethought from designing for physical products alone. Systems design is key for effectiveness for these product/service systems (PSSs) and the subject of this article.

PSSs have been studied, defined, and refined in scientific research, setting the practical stages for this article as a specific type of value cocreation system.

What Are Product/Service Systems (PSSs)?

PSS has been defined as "tangible products and intangible services designed and combined so that they jointly are capable of fulfilling specific customer needs" (Tischner *et al.* 2002). A well-known example is an offering by Rolls-Royce with aerospace engines named Power By The Hour, which include services in the offering and charge based on the number of hours for engine operation instead of the number of engines [Smith 2013].

The fundamental difference between providing a PSS and selling a product lies in the commitment by the provider—as depicted in Figure 1 (adapted from Baines *et al.* [2007]). The output from the product operation is often the target in PSS—the delivered product itself in product sales.

Integrating products and services in a packaged PSS offering is often designed to create higher customer value, such as hassle-free product operation or higher performance. It also has the potential to increase efficiency in terms of economy—both for the provider and the customer—and environment. Providing services separately in terms of contracts from selling the related products, which is traditional in many sectors, is simply wasteful. For instance, selling engines first and thereafter selling services when needed for the engines loses an opportunity for holistic optimization. In that traditional contractual relationship, the product seller is not economically incentivized to prolong the lifetime of the products.

We highlight some examples of PSS economic and environmental consequences. The first one is an example from a supplier of physical and digital visibility solutions sold to industrial customers for brand exposure [Kaddoura *et al.* 2019]. The company, Storbildsfabriken AB, Linköping, Sweden, is responsible for graphic design, arranging transportation for their customers, installation, and storage between uses. One of its products is an event tent. The product is typically used for one event only before its end of life, with the product sold to the user who disposes of it after its first use. In a PSS offering, the ownership of the product is instead retained by Storbildsfabriken, which leases it to the customer for an event and collects it.

Storbildsfabriken then refurbishes the product by reusing the aluminium frame and steel weights and replacing the textile, which is customized for each event. The refurbishing process also includes cleaning the frame and ensuring that

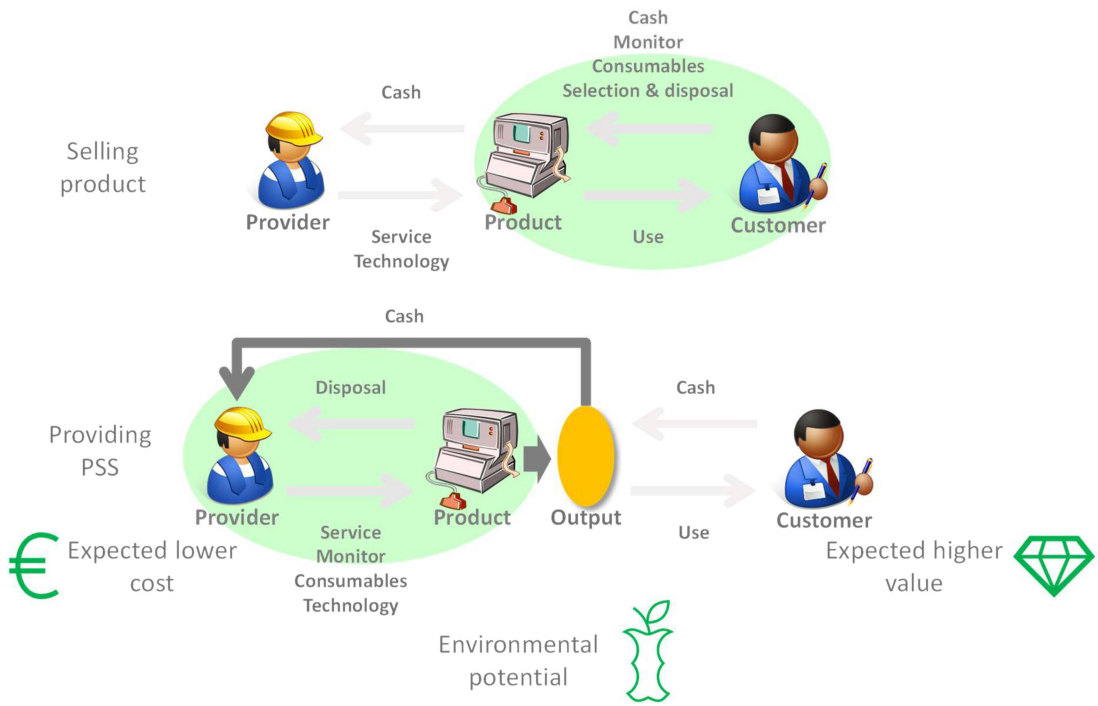


Figure 1. Comparison between providing PSS and selling product; adapted from Baines *et al.* [2007].

it is in good condition. The frame and weights are estimated to be used 10 times before the end of life in the PSS.

The comparison of selling the product versus PSS is depicted in Figure 2 in terms of economic cost (of the provider) and environmental impact (global warming potential in this graph). Figure 2 shows the cost and environmental impact total, normalized with the product sales, and the portions from different lifecycle phases for a single event.

The cost and the environmental impact in PSS are reduced by 18% and 45%, respectively. The graphs in Figure 2 also show how the contributions are shifted from product sales to PSS, both economically and environmentally.

The benefits of the PSS originate primarily from the savings in production—the grey portions with vertical lines; simply put, the production in the PSS per function is one-tenth the cost of the product environment. These savings outweigh the contributions by the newly introduced activities in the PSS, such as take-back (reverse logistics) and refurbishing (the green portions with diagonal lines). Value creation also differs—product sales relies on production, while the PSS on refurbishing. The main reason for the difference in the degrees of reduction between the cost and environmental impact is that the relative portions of the refurbishment are different – the portion is bigger in the cost than in the environmental impact in this PSS example.

Another example is introduced from a PSS with more complex products, soil compactors, based on our group's earlier research at Linköping University [Lindahl *et al.* 2014]. A soil compactor is primarily used to compact soil to create a strong base in construction sites.

The comparison carried out for the offerings with soil compactors by Swepac International AB (Swepac) is shown in Figure 3. The PSS includes remanufacturing (once) and maintenance by Swepac during the product life, and the lifetime is extended to eight years from five in the case of product sales. The product in the PSS is also designed differently from that in product sales: the varnishing paint has been replaced by galvanized steel for the PSS to increase longevity and avoid repainting. Thus, both the cost and environmental impact are superior in the PSS.

Note that the two companies presented here operate in Sweden, where the personnel cost is high even within the postindustrial highly technological economy. This implies the economic superiority of the PSSs, where the personnel cost is a major contributor is likely to be maintained.

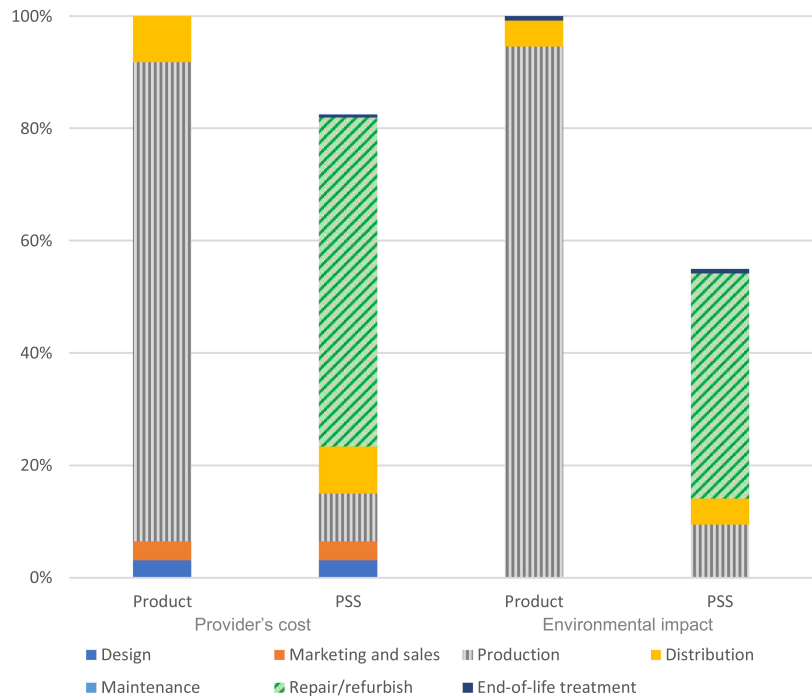


Figure 2. Comparison between selling a product (set to be 100%) and providing a PSS in provider's cost and environmental impact (global warming potential); modified from Kaddoura *et al.* [2019].

What Are the Implications of PSS Design?

The system to be designed is enlarged due to jointly including product and service, which has implications on design processes and activities. Design, especially design stages, is critical to effectively addressing this joint design environment.

Five essences for PSS design exist [Sakao and Neramballi 2020]. *Essence 1, development and integration of system elements*, and *Essence 2, examination of the balance of the integration*, are implied by PSS design systems thinking. Essence 2 exists because the product and service elements have *exchangeability* [Sakao and Lindahl 2015] in PSS design; meaning that efforts spent for the product and service can be integrated across both. For instance, refer to the example of the product produced and refurbish service in Figure 2.

These implications are further presented in Table 1, which summarizes the implications related

to the properties and characteristics of PSSs [Sakao *et al.* 2020]. Essences 1 and 2 are part of the open process systems property.

Essence 3, value propositions, and *Essence 4*, functionality-oriented designing, are most aligned with the business model property in Table 1. *Essence 5*, identification of relevant actors along the life cycle of PSSs, is implied by the social construct property in Table 1.

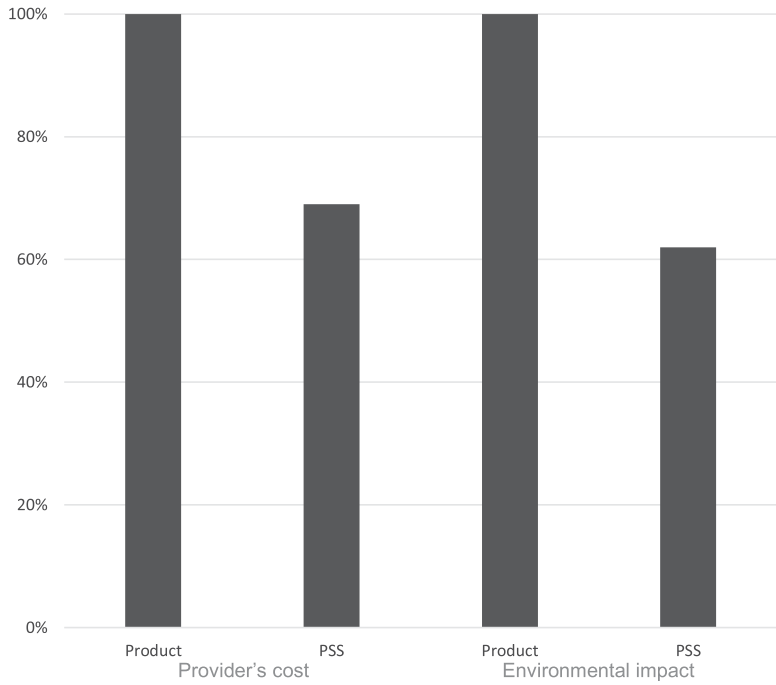


Figure 3. Comparison between selling a product (set to be 100%) and providing a PSS in provider's cost and environmental impact (global warming potential); modified from Lindahl *et al.* [2014].

How Can PSS Design Be Implemented in a Manufacturing Company?

To implement PSS design using the five essences for complex products requires guidance for designers and managers. An example for the guidance is represented in the form of a prescriptive PSS design schema, as shown by Figure 4 [Sakao and Neramballi 2020].

This process may appear overwhelming. However, this schema was developed by pouring the five essences of PSS design into the mold of a widely used systematic approach to engineering design by Pahl's and Beitz's framework [Pahl *et al.* 2006] (see Table 2).

Therefore, it is similar to standard design procedures adopted in many manufacturing companies for physical products and is easily adapted from existing procedures for PSS design. The schema is a practical guide showing aspects to consider during conceptual PSS design processes.

Table 1. Key PSS Properties and Characteristics and Their Implications on Its Conceptual Design [Sakao* et al. *2020]

Property	Characteristic	Implication for conceptual design of PSS
Open process systems	Human activities Heterogeneity Uncertainty System architecture System components System behavior Inputs and outputs Processes and functions	Apply systems thinking. Integrate the product and service views. Analyze behavior as a system. Consider uncertainty.
Business model	Nature of business Value proposed Customer orientation Performance of asset Available resources	Consider business model. Analyze customers. Include value proposition. Consider system performance. Consider service personnel.
Social construct	Actors' roles and scenarios Technological and socio-cultural interactions Relationship between customer and provider	Analyze actors' roles. Analyze scenarios. Apply cocreation process.

Note: The first property, open process systems, means that a PSS is a system with input and output flows in the following sense: Output flows are determined by processes and functions in the PSS, which involve human activities characterized by heterogeneity inherited from the generic characteristics of pure service. The processes in PSSs also involve product behaviors that change over time due to, for example, deterioration. The service heterogeneity and the product change over time are both uncertain. The next property is the business model, which takes into account the nature of the businesses involved in the products and services. The last PSS property, social construct, means that more actors are involved in terms of roles and scenarios than in product sales. For instance, technological and sociocultural interactions are relevant [Morelli 2003]. Furthermore, Baines *et al.* [2007] assert that the relationship between the customer and provider is an important characteristic. References given in this article are omitted from the table due to the increased readability.

Some of the 10 steps are peculiar to conceptual PSS designing, in contrast to Pahl's and Beitz's framework. *Essence 1. Functionality-oriented designing* is suitable to Steps 1 and 2, whereas *Essence 5. Examination of the balance of the integration* is most applicable to Steps 5 and 6.

Note that this schema represents what was reported to have worked in cases and thus is based on actual practice. How experienced designers would perform PSS design is a different, descriptive question; our ongoing research is trying to answer if it is different from product design and, if so, how [Neramballi *et al.* 2019].

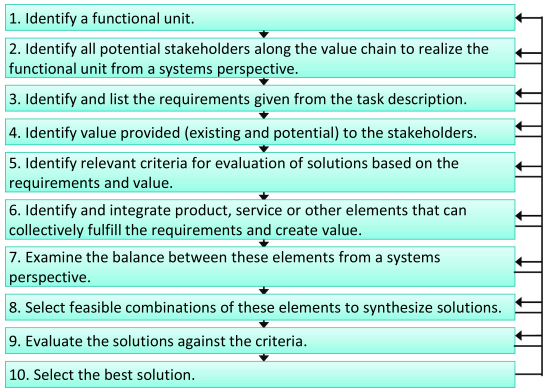


Figure 4. Ten-step PSS design schema [Sakao and Neramballi 2020].

Companies with a strong legacy in technology innovation and product design might find designing services unfamiliar. Therefore, *what* to design could be worthwhile to be guidance for services design. I have yet to meet a staff member in a manufacturing organization in a service designer (or service developer) position—in stark contrast to millions of product designers or product developers.

Before services can be provided, they should be designed—the degrees of detail and care do differ between cases. In PSS design, services as part of systemic design need to be carefully considered taking a holistic perspective of all product and service elements taken into account.

Table 3 shows a generic list of service design characteristics (i.e., design

Table 2. Representation of the Consolidation of PSS Design Schema [Sakao and Neramballi 2020]

Pahl's and Beitz's Framework for Conceptual Design	Distilled Essence of PSS Design	Consolidated Schema for Conceptual PSS Design
1. Abstract to identify the essential problems 2. Establish function structures: overall function—subfunctions	1. Functionality-oriented designing 2. Identification of relevant actors along the PSS lifecycle	1. Functional unit definition 2. Stakeholder identification 3. Requirement consolidation
3. Search for working principles that fulfill the subfunctions 4. Combine working principles into working structures	3. Value proposition 4. Development and integration of system elements	4. Value proposition 5. Criterion identification 6. Element integration
5. Select suitable combinations 6. Firm up into principle solution variants	5. Examination of the balance of the integration	7. Balance examination 8. Selecting combinations
7. Evaluate variants against technical and economic criteria	No essence applicable	9. Evaluating combinations 10. Solution selection

Table 3. Service Design Characteristics and Their Measurables [Neramballi* et al. *2020]

Service Design Characteristic	Description of Measurables
Responsiveness	*Time* taken to deliver a service/response to the customer.
Labor intensity	Level of involvement of human resource in a service activity. Measured in terms of the *number* of personnel involved in the activity and the *effort* spent by the personnel.
Customizability	Ability to provide a customer the options to personalize the offering based on their needs. Measured in terms of *quantity* and *type* of information collected from customers (input) and corresponding *choices* given to personalize the offering (output).
Affordability	Level of pricing for a service activity in *monetary* terms.
Security	Level of protection of the customer data/information. Measured in terms of security of user information, security while using service, transparency, confidentiality, etc.
Reliability	Level of dependability of fulfillment of the intended functions of service activity. Measured in terms of *consistency of accurate results* of service activities.
Modularity	Level to which offerings can be separated and recombined based on customer needs. Measured in terms of the *absolute number* of potential modular alternatives of offerings.

parameters) based on a review of the literature that has described service characteristics of various industrial service offerings [Neramballi *et al.* 2020]. This list is not meant to be exhaustive but can be utilized as a guiding tool while deriving the service design parameters for a PSS in question.

The time dimension for service design parameters is necessary because product and service design parameters substantially differ in terms of lifecycle time windows [Sakao *et al.* 2022].

Service is typically inseparable between production and use; that is, whereas products are produced and subsequently used, services are produced and consumed simultaneously. The consumption of products and services occurs at any point after the offering is deployed to the user—design must be completed before their associated production.

Thus, a contrast emerges between products and services about when design, production, and use take place, as depicted in Figure 5. Before the start of product use (consumption), design and then production must occur for products, whereas only design can be carried out for services. After the start of product use, its design is generally frozen, whereas services can still be designed as a result of its inseparability (or simultaneity) principle. Thus, higher flexibility is afforded when service design is completed.

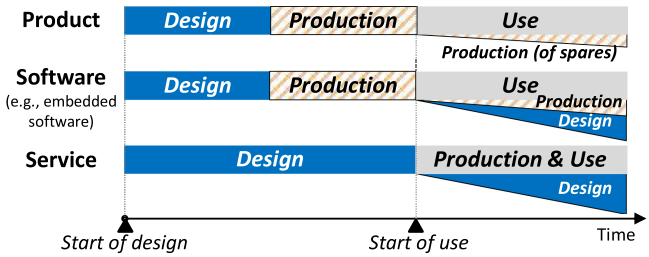


Figure 5. Possible time windows to design product, software, and service [Sakao *et al.* 2022]. Note: This figure is simplified, as concurrent design and production of products as well as product upgrade are not covered in this article.

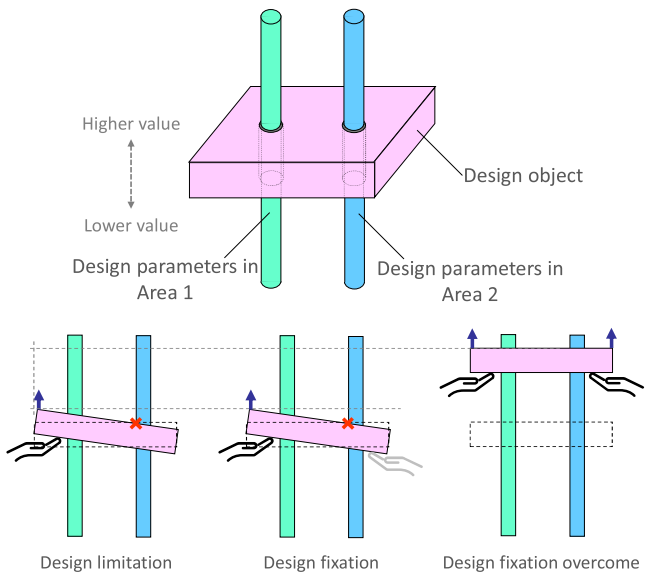


Figure 6. Distinction between limitation and fixation from the perspective of designing [Wasserbaur and Sakao 2020] developed from Sakao and Shimomura [2007]. Note: Suppose the design object in question has two sets of design parameters, e.g., in the product area and the service area. A design limitation exists when the designer has control only in one of the areas. When the designer has control in both areas but does not use it in Area 2, fixation is observed. When the control in both areas is actually used effectively, the design fixation is overcome.

This higher flexibility on the temporal dimension allows PSS designers to postpone service design in part (i.e., not entirely) and thereby effectively and efficiently address the uncertainty arising in the use phase; which grants a competitive advantage to a manufacturer as a PSS provider by offering greater flexibility for differentiation.

Digitalization provides increasing possibilities and impacts of collecting and utilizing data for efficient PSS provision. The question, however, is which service design aspects should be decided upon in integration with the product design and which aspects ought to be postponed to the use phase. To answer this question, understanding what information and intelligence will be obtained in the use phase is crucial—something that digitalization increasingly offers ([Sakao and Nordholm 2021]; [Sakao *et al.* 2021]).

Practical Lessons Learned From Early PSS Design Research

Implementing PSS design can be described effectively, but it is deeply challenging to practically realize in large and complex organizations. Let me share some lessons learned through collaboration with large global manufacturers implementing PSS design.

Some of the five essences may be depicted as a metaphor, as shown in Figure 6. The PSS provider wants to

increase the value of the PSS (refer to *Essence 3*) by heightening the design object. Various organizational functions and departments provide relevant design parameters—for example the design parameters emanating from "Area 1" and separately from "Area 2" in Figure 6. Hence, coordinating between the areas and integrating the parameters from those is needed to increase the value effectively [Sakao and Shimomura 2007], corresponding to *Essences 1 and 2*.

Organizational challenges typically exist because of tradition, cultural mindsets, power balance, and other intangible organizational characteristics. Addressing the interplays between products and services is challenging and often a lost or unrecognized opportunity due to these environmental factors.

This unexploited potential may be due to something defined as *design fixation*, which refers to blind adherence to a set of existing concepts. Design fixation limits the output of the conceptual design process [Jansson and Smith 1991] also at the organizational level [Wasserbaur and Sakao 2020].

One PSS provider often failed to use the service design parameters in integration with product design parameters, adhering to existing product-focused design solutions—see the bottom middle of Figure 6. Here, it is important to distinguish between design fixations and limitations. The causes of a design fixation are controllable by the designer, whereas those of a design limitation are not, for example, natural laws and societal regulations. The usefulness of this limitation–fixation dichotomy lies in its potential to increase creativity. A designer ought to carefully investigate whether they blindly adhere to known concepts. Once a design parameter is identified to be an object of design fixation, better solutions can be found—the bottom right of Figure 6.

The timing, the "when," to address service design parameters in detail is an important PSS design decision—as visualized in Figure 5 [Sakao *et al.* 2022]. From our studies and experiences, the opportunity of partly postponing service design is often not optimally seized in practice by manufacturers. Namely, too many or too few decisions are made for the service design at the end of product design. An example is assigning excessive resources such as reserving service technicians and shipped spare parts fixed to the service, whereas not clearly planning for an overhaul.

This type of suboptimal design causes substantial negative effects on the entire PSS. These issues present the risk of a lost opportunity for the optimal or better PSS design. To efficiently communicate this risk we coined a term *service design trap*, which refers to a phenomenon where the ability of service designers over an extended timespan (shown in Figure 5) is restricted due to the constraints set by the frozen product design during the use phase. The phenomenon is visualized in Figure 7, which further identifies the risks associated with the service design trap.

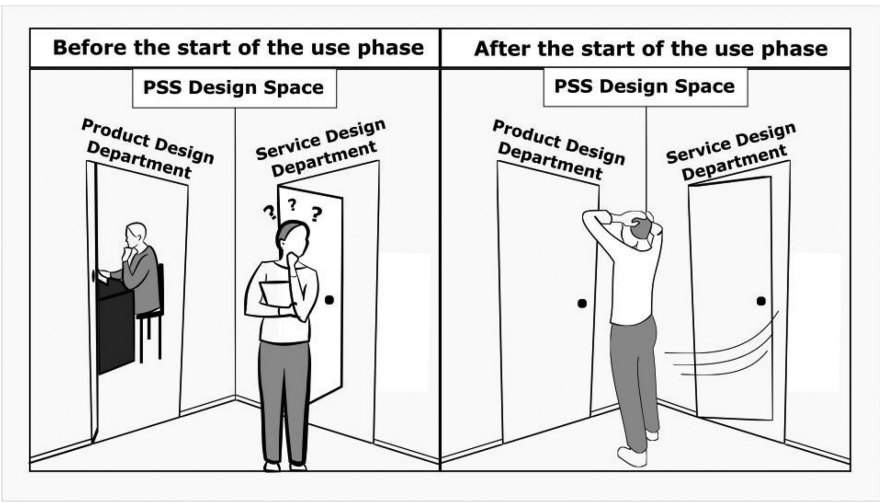


Figure 7. Visualization of the service design trap [Sakao *et al.* 2022]. Note: A service designer is standing in the PSS design space. The service designer has access to both the product and service design spaces, meaning the possibility to interact and influence the design at the product and service design departments. This possibility is present only before the use phase. During the use phase, the product design is frozen, which may cause negative impacts on PSS performance. The service designer has become trapped on the right-hand side after he or she enjoyed the higher freedom for the service design inappropriately on the left-hand side.

Addressing organizational challenges requires the top-management commitment to PSS. Only with such commitment is the PSS design process, we envisage in this article, could be possible. One potential solution for management in implementing PSS design is designating a PSS facilitator to coordinate integration activities within the company [Matschewsky *et al.* 2018].

SUMMARY

My hope is that this article attracts the interest of an existing or potential PSS development in organizations. We provide an understanding of the *what, why, and how* for PSS design. Some of the lessons learned and imparted here can facilitate PSS design in the future.

PSS design is the planning of value cocreation, which is often unique to any company. Therefore, each PSS provider must establish its way of PSS design. Finally, we can summarize the PSS design explained in this article with seven "*I*s" that can help a reader recall its major elements including the five essences:

- *Integrative*—integration between products and services comes first in importance of all the aspects. To be specific, use interplay, interdependence, interference, and exchangeability between products and services.
- *Information driven*—take advantage of the availability of data/information from different phases of the product lifecycle. When in design, using the relevant information is key for successful service design to avoid the service design trap, as illustrated in Figure 7.
- *Intelligent*—build upon business intelligence (not only artificial intelligence) to make appropriate value propositions in PSS design. Remember potential contributions by digitalization.
- *Interactive*—use the interaction between users and PSSs (and the provider), not only that between products and services. One of the PSS properties is the social construct (as shown in Table 1).
- *Inclusive*—carefully consider the inclusion of other (new) necessary actors involved in the enlarged system (not only the payer, the user, and the producer). Remember Essence 5 *for PSS design, Identification of relevant* actors.
- *Indefinite*—reflect the uncertainty of the PSS, its inherent characteristic, and its advantage. Uncertainty of products and services during the use phase is key, as shown in Figure 5.
- *Innovative*—avoid fixation in PSS design; recall Figure 6.

As a manufacturing company, you may want to begin PSS design as a project and familiarize your staff with it. Then, you will face other challenges regarding how to manage the transformation process in your organization, advancing the services provided to your products. We will follow up with these transformation processes in a future issue of IEEE Engineering Management Review. Note that a reader interested in an academic, extensive state-of-the-art can read a review article [Brissaud *et al.* 2022].

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References

Baines, T. S., H. W. Lightfoot, S. Evans, A. Neely, R. Greenough, J. Peppard, R. Roy, E. Shehab, A. Braganza, A. Tiwari, J. R. Alcock, J. P. Angus, M. Bastl, A. Cousens, P. Irving, M. Johnson, J. Kingston, H. Lockett, V. Martinez, P. Michele, D. Tranfield, I. M. Walton and H. Wilson (2007). "State-of-the-art in product-service systems." *Proceedings of the Institution of Mechanical Engineers - B* 221: 1543–1552.

Brissaud, D., T. Sakao, A. Riel and J. A. Erkoyuncu (2022). "Designing value-driven solutions: The evolution of industrial product-service systems " *CIRP Annals - Manufacturing Technology* 71.

Jansson, D. G. and S. M. Smith (1991). "Design fixation." *Design Studies* 12(1): 3–11.

Kaddoura, M., M. L. Kambanou, A.-M. Tillman and T. Sakao (2019). "Is prolonging the lifetime of passive durable products a low-hanging fruit of a circular economy? A Multiple case study." *Sustainability* 11, Art. no. 18.

Lindahl, M., E. Sundin and T. Sakao (2014). "Environmental and economic benefits of integrated product service offerings quantified with real business cases." *Journal of Cleaner Production* 64: 288–296.

Matschewsky, J., M. L. Kambanou and T. Sakao (2018). "Designing and providing integrated product-service systems – challenges, opportunities and solutions resulting from prescriptive approaches in two industrial companies." *International Journal of Production Research* 56(6): 2150–2168.

Morelli, N. (2003). “Product-service systems, a perspective shift for designers: A case study: The design of a telecentre.” *Design Studies* 24(1): 73–99.
Neramballi, A., T. Sakao and J. S. Gero (2019). What do experienced practitioners discuss when designing product/service systems? *Design Computing and Cognition ’18*, J. S. Gero, Ed. New York, NY, USA: Springer**,** 361–380.
Neramballi, A., T. Sakao, S. Willskytt and A.-M. Tillman (2020). “A design navigator to guide the transition towards environmentally benign product/service systems based on LCA results.” *Journal of Cleaner Production* 277, Art. no. 124074.
Pahl, G., W. Beitz, J. Feldhusen, K. H. Grote, K. Wallace and L. T. M. Blessing (2006). *Engineering Design: A Systematic Approach*. London, U.K.: Springer.
Sakao, T., P. Funk, J. Matschewsky, M. Bengtsson, and M. Uddin Ahmed (2021). “AI-LCE: Adaptive and intelligent life cycle engineering by applying digitalization and AI methods – An emerging paradigm shift in life cycle engineering.” *28th CIRP Conference on Life Cycle Engineering*: 571–576.
Sakao, T., J. Gero and H. Mizuyama (2020). “Analyzing cognitive processes of a product/service-system design session using protocol analysis.” *Artificial Intelligence for Engineering Design, Analysis and Manufacturing* 34(4): 515–530.
Sakao, T. and M. Lindahl (2015). “A method to improve integrated product service offerings based on life cycle costing.” *CIRP Annals - Manufacturing Technology* 64(1): 33–36.
Sakao, T. and A. Neramballi (2020). “A product/service system design schema: Application to big data analytics.” *Sustainability* 12, Art. no. 3484.
Sakao, T., A. Neramballi and J. Matschewsky (2022). “Avoid service design trap by guiding product/service system design with product-service dependency knowledge base.” *International Design Conference*.
Sakao, T. and A. K. Nordholm (2021). “Requirements for a product lifecycle management system using Internet of Things and big data analytics for product-as-a-service.” *Frontiers in Sustainability*, 2, Art. no. 735550.
Sakao, T. and Y. Shimomura (2007). “Service engineering: A novel engineering discipline for producers to increase value combining service and product.” *Journal of Cleaner Production* 15(6): 590–604.
Sakao, T. and K. Webster. (2020). “The wash cycle – Benefits of product-as-a-service in a circular economy,” Ellen MacArthur Foundation, Cowes, U.K.[Online]. Available: https://medium.com/circulatenews/the-wash-cycle-6aaed1a2c310
Smith, D. J. (2013). “Power-by-the-hour: The role of technology in reshaping business strategy at Rolls-Royce.” *Technology Analysis & Strategic Management* 25(8): 987–1007.
Tischner, U., M. Verkuijl and A. Tukker (2002). “First Draft PSS Review,” Econcept, Cologne, Germany.
Wasserbaur, R. and T. Sakao (2020). “Conceptualising Design Fixation and Design Limitation and Quantifying Their Impacts on Resource Use and Carbon Emissions.” *Sustainability* 12(19), Art. no. 8104.

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